

NEET Important Chapter – Moving Charges and Magnetism

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The chapter moving charges and magnetism deals with the magnetic field produced by the current-carrying wire and the magnetic force acting on a current-carrying wire and moving charge in a magnetic field. As far as the NEET exam is concerned, problems will be asked related to concepts discussed in the chapter.

The chapter moving charges and magnetism begins with the magnetic force acting on a charged particle moving in a magnetic field. Cyclotron which is a practical application of Lorentz force is discussed in this chapter in detail. Biot-Savart law and Ampere's law are very important concepts in this chapter.

We also learn about the formula of magnetic effect of electric current and magnetic field produced due to a solenoid and toroid. Force acting on a current-carrying conductor placed in a magnetic field and the force acting between two parallel straight conductors are also discussed in this chapter.

Now, let us move on to the important concepts and moving charges and magnetism formulas related to the [NEET 2022](#) exam from the chapter moving charge and magnetism along with a few solved examples.

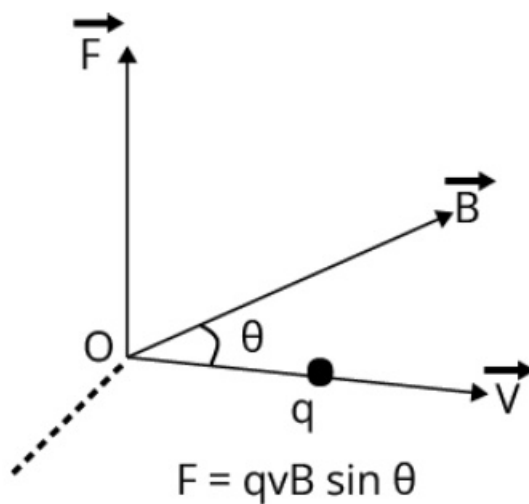
Important Topics of Moving Charges and Magnetism

- Magnetic force on a moving charge in a magnetic field
- Trajectory of a charge moving in a perpendicular magnetic field
- Cyclotron
- Biot Savart Law
- Ampere's Law
- Magnetic field due to a circular and straight wire
- Magnetic field due to a solenoid
- Force on a current-carrying conductor in a magnetic field.
- Force between two parallel current-carrying conductors

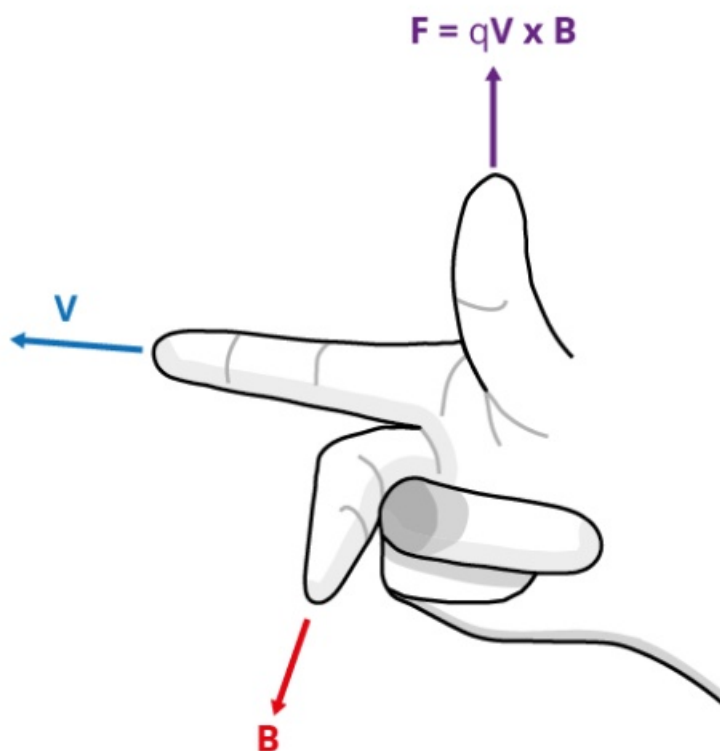
Important Concepts of Moving Charges and Magnetism

Name of the Concept	Key Points of the Concept
	• A moving charge produces an electric field and magnetic field. • The charge (q) moving with a velocity (v) in a magnetic field (B) experiences a magnetic force(F) given by

1. Magnetic force on a moving charge in a magnetic field

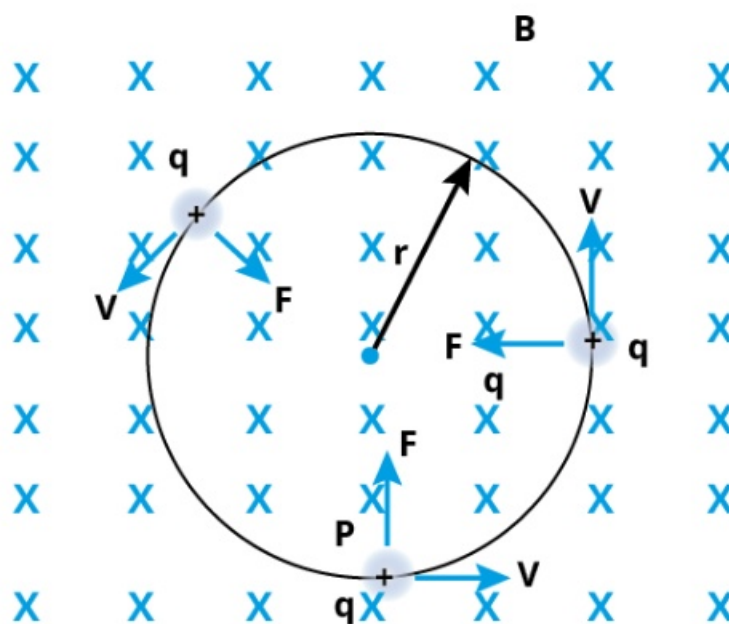


- The direction of magnetic force is given by right-hand rule as shown below



2. The trajectory of a charge moving in a perpendicular magnetic field

- When a charge (q) moves in a perpendicular magnetic field (B), then it follows a circular path and necessary centripetal force is given magnetic force.



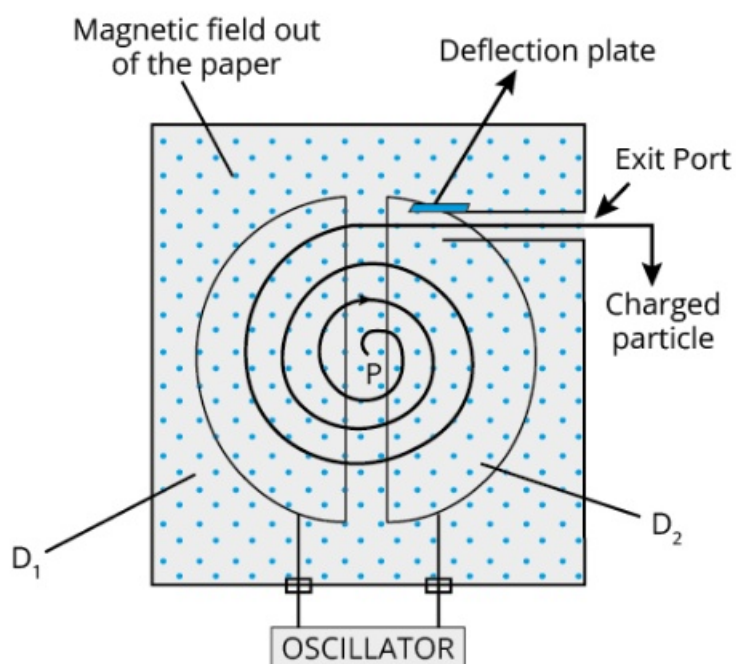
- The radius (r) of the circular path is given by

- Time period of motion of charge in a circular path is

- Kinetic energy of circular path is given by

3. Cyclotron

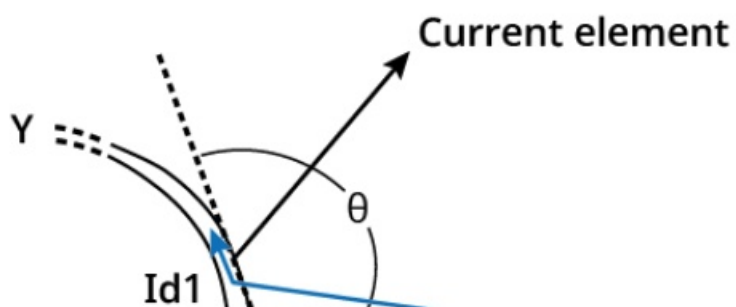
- Cyclotron is a device used to accelerate positively charged particles.

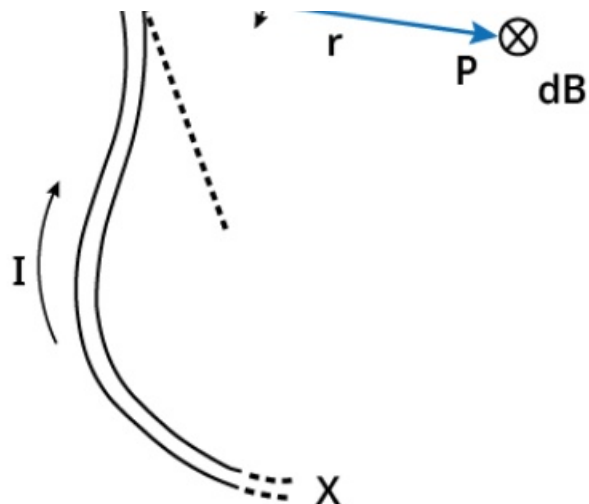


- The charged particles are accelerated by an alternating electric field created by two D shaped semicircular hollow chambers in a perpendicular magnetic field.

- Since the charged particle is moving in a perpendicular magnetic field, it follows a circular path of increasing radius
- The cyclotron frequency is given by

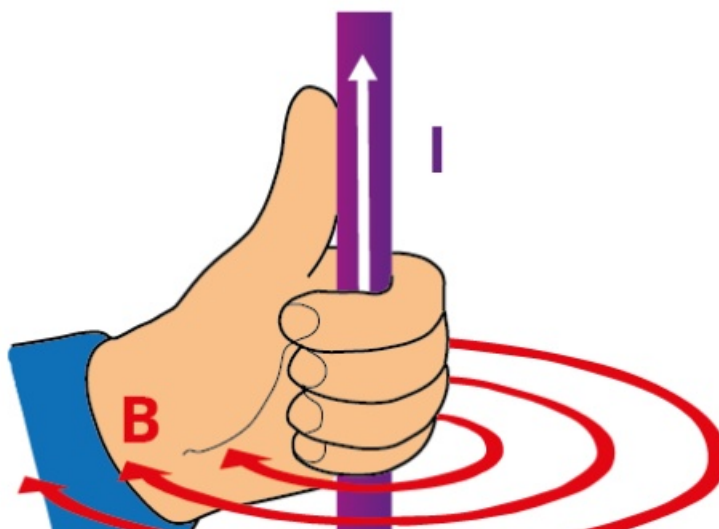
- According to Biot Savart law, the magnetic field (dB) at a point due to the current element (dl) is given by the expression





Where μ is the magnetic permeability of the medium and for free space, magnetic permeability $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

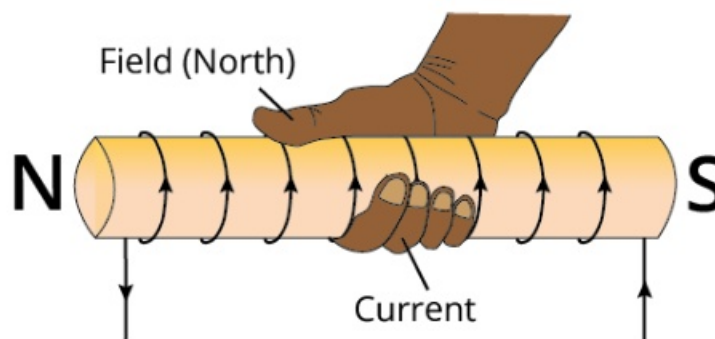
- If you hold a current-carrying conductor in such a way that your thumb points towards the direction of current, then the remaining fingers give the direction of the magnetic field at that point.



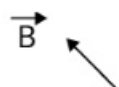
4. Biot Savart Law



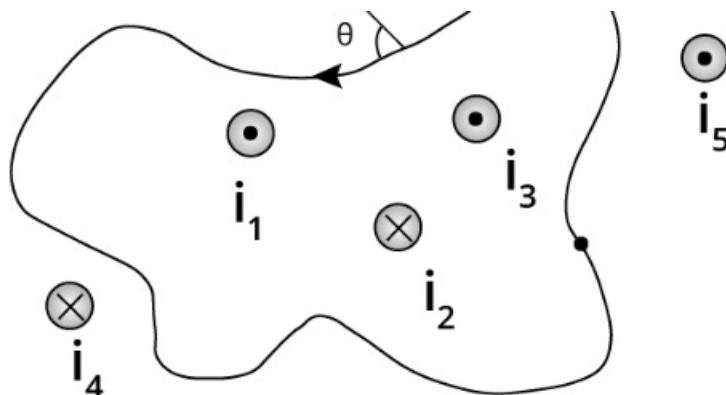
- In the case of a circular current-carrying coil, if the four folding fingers of the right-hand point towards the direction of the current, then the thumb point towards the direction of magnetic field



- Ampere's law gives a method to calculate the magnetic field due to a given current distribution.
- According to ampere's law, the line integral of the magnetic field around any closed curve is equal to μ_0 times the net current enclosed by the closed curve.



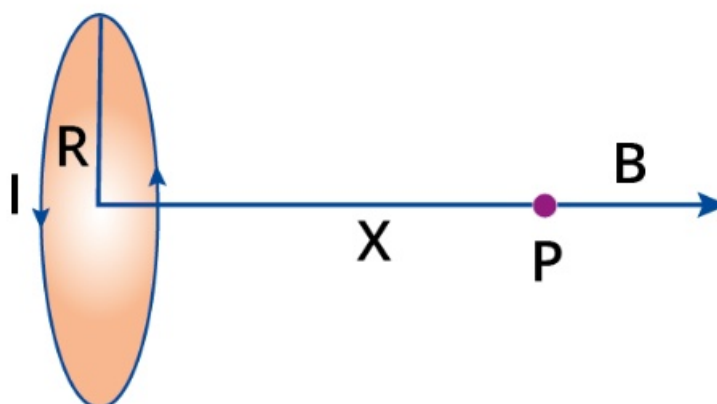
5. Ampere's Law



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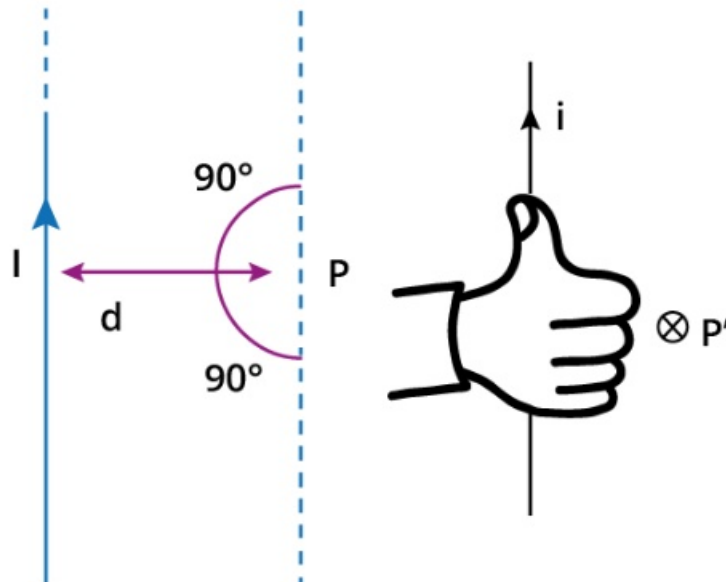
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- Magnetic field due to a circular coil having current I and radius R at a distance x along its axis is given by



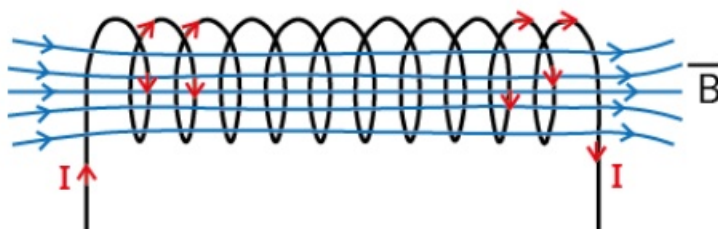
6. Magnetic field due to a circular and straight wire

- Magnetic field due to an infinitely long current-carrying wire having current I at a distance d is



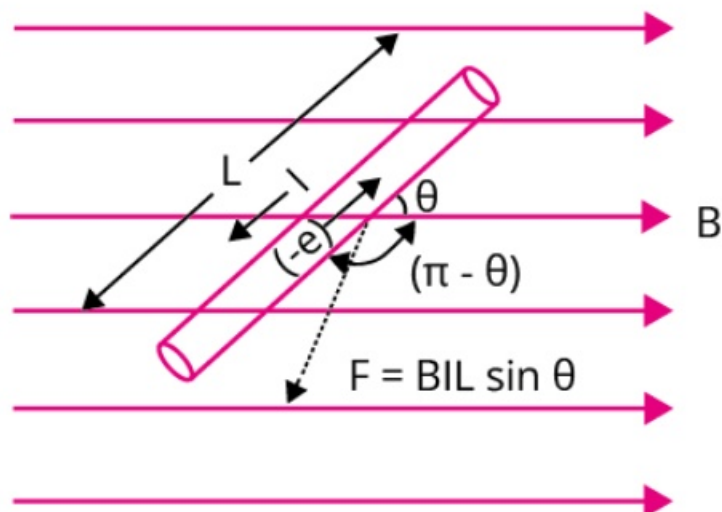
7. Magnetic field due to a solenoid

- A solenoid is a tightly wound cylindrical coil of insulated wire having a smaller diameter than its length
- Magnetic field due to a solenoid having n number of turns per unit length along its axis outside the solenoid



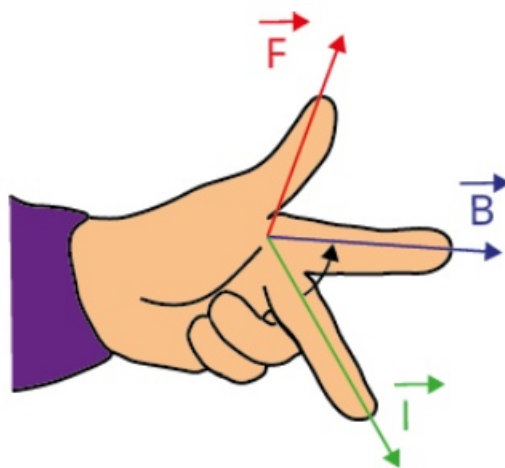
- When a current-carrying conductor having length L is placed in a magnetic field B , it will experience a force given by

the formula,



8. Force on a current-carrying conductor in a magnetic field.

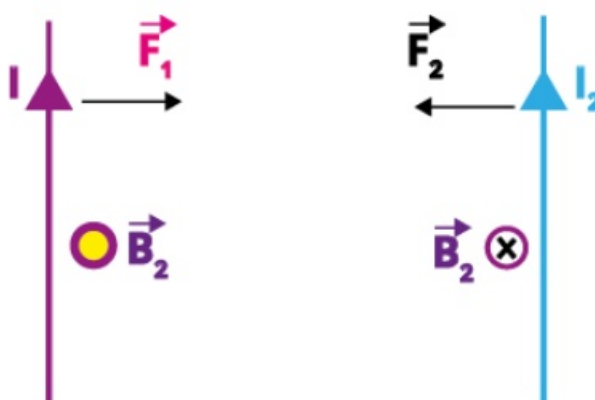
The direction of force is given by Fleming's left-hand rule as given below



9. Force between two parallel current-carrying conductors

- Force acting between two parallel current-carrying conductors having length l is given by

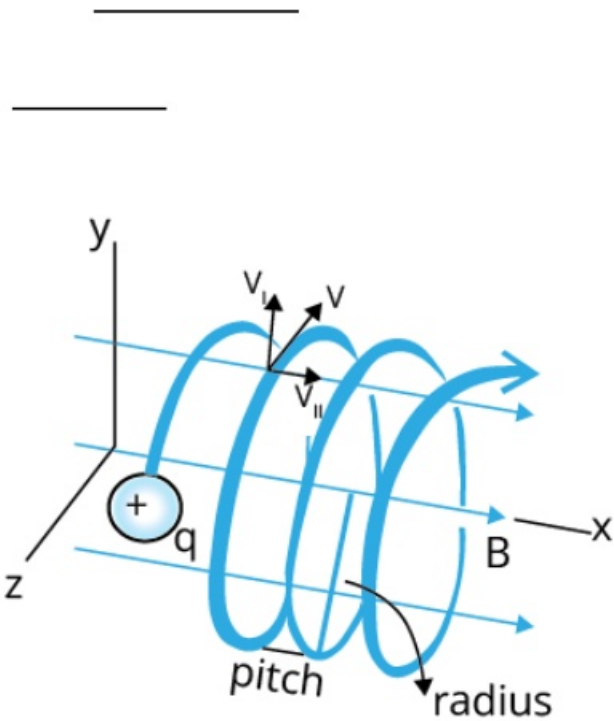
$$\frac{\mu_0}{4\pi} \frac{I_1 I_2 l}{r}$$

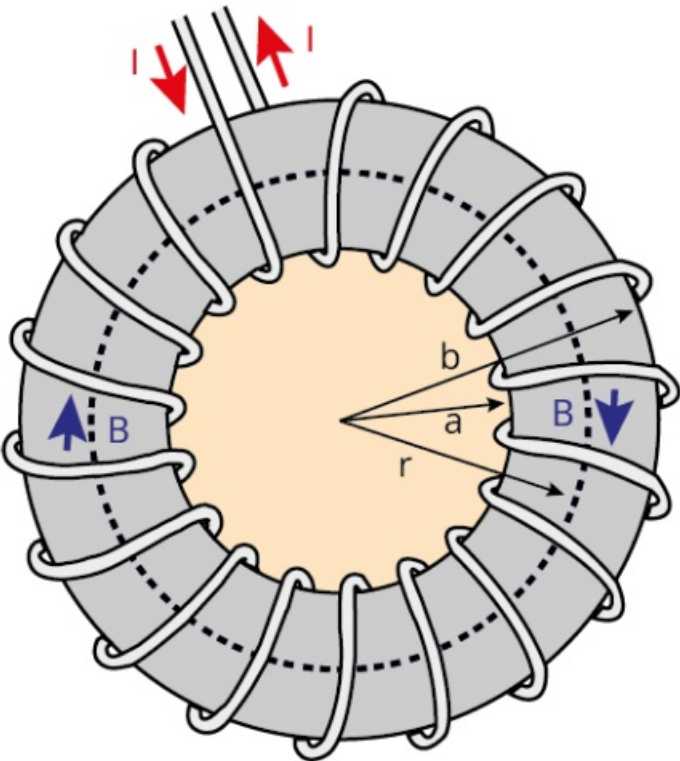
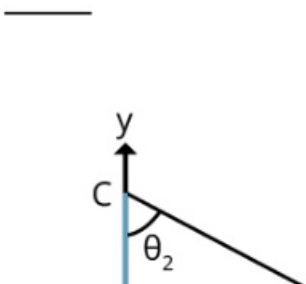


Current carrying conductors

- If the current in the conductors is in the same direction, there will be the force of attraction and if the currents are in the opposite direction, then there will be the force of repulsion

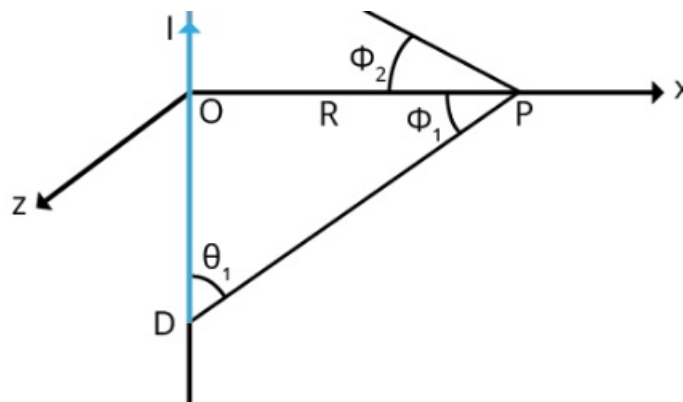
List of Important Formulae

Sl. No	Name of the Concept	Formula
1.	Lorentz force formula	The force acting on a charged particle (q) in a combined electric field (E) and magnetic field (B)
2.	Pitch and radius of a helical path of a charged particle (q) moving at an angle θ to the magnetic field (B) with a velocity (v)	 The diagram illustrates a charged particle (q) moving in a helical path within a uniform magnetic field (B). The magnetic field is represented by blue arrows pointing along the positive x-axis. The particle's velocity vector (v) is shown at an angle θ to the x-axis. This velocity is decomposed into a component $v_{\parallel}$ parallel to the magnetic field and a component $v_{\perp}$ perpendicular to it. The perpendicular component causes the particle to move in a circular path with a radius, while the parallel component causes it to move along the x-axis, resulting in a helical path. The distance between two consecutive loops of the helix is labeled as the pitch. A 3D coordinate system with x, y, and z axes is shown for reference.

3.	Magnetic field inside a toroid	
4.	The magnetic moment of a current-carrying coil having an area of cross-section (A) and N number of turns	
5.	The torque acting on a current-carrying coil placed in a magnetic field B	Where θ is the angle between area vector and magnetic field and N is the number of turns
	The magnetic field at	

6.

a point which lies on the perpendicular bisector of the finite length of straight wire



Solved Examples

1. Two long parallel wires carrying equal current separated by 1 m, exert a force of $2 \times 10^{-7} \text{ N/m}$ on one another. The current flowing through them is

Sol:

Let the Current passing through each wire be I

The force per unit length acting on each wire is given as $2 \times 10^{-7} \text{ N/m}$

The formula to calculate the force exerted by wires on one another is given by,

Therefore, the total current flowing through them is 1 A.

Key Point: Force acting between two parallel current-carrying wires depend on the current passing through the wires and distance between them.

2. An alpha particle and proton travel with same velocity in a magnetic field

perpendicular to the direction of their velocities, find the ratio of the radii of their circular path.

Sol:

Let the mass of the proton and alpha particle be 'm' and '4m' respectively. The charge of the alpha particle is two times the charge of the proton.

The formula to calculate the radius of the circular path of proton is given by

$$r = \frac{mv}{qB} \dots(1)$$

The radius of the circular path of the alpha particle is given by,

$$r = \frac{mv}{qB}$$

$$r = \frac{4mv}{2qB} \dots(2)$$

Divide equation (1) by equation (2) to obtain the ratio of radii of their circular path.

$$\frac{r}{r} = \frac{\frac{mv}{qB}}{\frac{4mv}{2qB}}$$

Therefore, the ratio of the radii of their circular path is 1:2

Key Point: Alpha particle is a He^{2+} nucleus and has mass four times the mass of a proton and charge equal to double the charge of proton

Previous Year Questions from NEET Paper

1. A 250 turn rectangular coil of length 2.1 cm and width 1.25 cm carries a current of 85 μA and subjected to magnetic field of strength 0.85 T. Work done for rotating the coil by 180° against the torque is (NEET 2017)

a. 4.55 μJ

b. $2.3 \mu\text{J}$

c. $1.15 \mu\text{J}$

d. $9.1 \mu\text{J}$

Sol:

The magnetic moment of the rectangular coil is calculated by the formula given below,

Work done in rotating the rectangular coil can be calculated as follows,

Therefore, correct option is option (d)

Trick: Magnetic moment of rectangular coil is equal to the product of number of turns, area of the coil and the current passing through the coil.

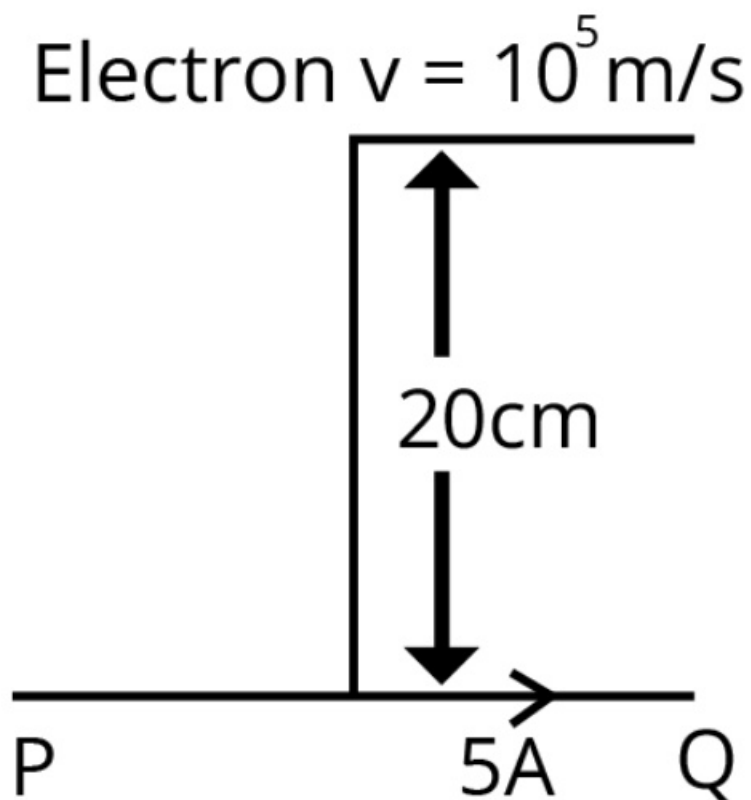
2. An infinitely long straight conductor carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant. (NEET 2021)

a. 4×10^{-20} N

b. $8\pi \times 10^{-20}$ N

c. $4\pi \times 10^{-20}$ N

d. 8×10^{-20} N



Sol:

The magnetic field due to current carrying infinitely long straight conductor is given by,

The force acting on the electron is given by

The correct answer is option (d).

Trick: The motion of a charged particle and the magnetic field produced by the infinitely long conducting wire is perpendicular to each other.

Practice Questions

1. A closely wound flat circular coil of 25 turns of wire has diameter of 10 cm which carries a current of 4 A. The flux density at the centre of the coil will be ? (**Ans: $1.256 \times 10^{-3} \text{ T}$**)
2. An electron is moving in a circular orbit of radius r makes n rotations per second. The magnetic field produced at the centre has magnitude _____ (**Ans: $\mu_0 ne/2r$**)

Conclusion

In this article we have provided important information regarding the chapter moving charges and magnetism such as important concepts, formulae, etc.. Students should work on more solved examples along with previous year question papers for scoring good grades in the NEET exam.

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